MAHMOUD ABD EL-GHAFFAR

AI Automation Engineer & Systems Integration Specialist

Make | Zapier | n8n | REST APIs | HubSpot & Airtable | Multi-Agent Workflows

Location: Benha / Cairo, Egypt • Phone: +20 102 419 0741 • Email: mahmoudzanaty454@gmail.com
LinkedIn: linkedin.com/in/mahmoud-abd-el-gaffar-4aa378219 • Portfolio: https://portfolio-mrt6.onrender.com/

PROFESSIONAL SUMMARY

Accomplished Computer Engineer and AI Automation Specialist with hands-on expertise building production workflows, multi-agent pipelines, and cloud integrations across Make, Zapier, n8n, and custom REST APIs. Built 30+ native integration nodes inside AI MicroMind and engineered 30+ business workflows connecting CRMs (HubSpot, Airtable), web forms, messaging platforms (Slack, Gmail, WhatsApp/WAPA), and databases. Known for systems thinking, proactive error handling, and explaining technical workflows in plain language to non-technical business clients. Comfortable working under strict NDA.

CORE TECHNICAL COMPETENCIES

- **Automation Platforms & APIs:** Make (Integromat), Zapier, n8n, Webhooks, Cron Schedulers, REST APIs, OAuth2, Postman.
- **CRMs & Business Tools:** HubSpot, Airtable, Gmail, Microsoft 365, Slack, Meta/WhatsApp (WAPA), Telegram, ClickUp, Typeform.
- **AI & Multi-Agent Frameworks:** LangChain, LangGraph, Model Context Protocol (MCP), Pinecone Vector DB, OpenAI, LLM-to-SQL.
- **Systems Reliability & Delivery:** Automated Error Handling, Fallback Triggers, Retry Logic, SOPs & Plain-Language Client Documentation.
- **Backend & Cloud Stack:** Node.js, TypeScript, Python, Docker, AWS, DigitalOcean Linux VPS, Render, GitHub CI/CD.

PROFESSIONAL EXPERIENCE

AI & Automation Engineer — AI MicroMind
2024 – Present

- **Core Integration Nodes:** Built 30+ native integration nodes inside AI MicroMind Core (Node.js) for WAPA, Telegram, Gmail, Pinecone, Webhooks, Schedulers, and PostgreSQL Audit Loggers using AI tools (Antigravity, Claude Code).
- **Business Workflows:** Designed multi-step automations across Zapier, Make, and n8n connecting CRMs (HubSpot, Airtable), web forms, and messaging apps, eliminating manual data entry.
- **Report Bot & BI:** Architected the AI MicroMind Report Bot Framework, enabling non-technical users to query on-premise ERPs (Oracle, SAP, Dynamics 365, Odoo) and SQL databases via plain language.
- **Error Handling & Reliability:** Implemented robust fallback triggers, alert notifications, and webhook retry mechanisms, ensuring zero data loss and 99.9% uptime.
- **Client SOPs & Handover:** Authored visual workflow diagrams and plain-language documentation to train and onboard non-technical business clients.

Automation Engineer — Operatives (IT Services)
Jun 2021 – Jan 2022

- **Workflow Automation:** Engineered cross-departmental automated synchronization pipelines, eliminating manual bottlenecks and cutting data errors by 70%.

Operation Engineer & Admin — Operatives (IT Services)
Jan 2020 – Jun 2021

- **Operations & SLA Compliance:** Administered IT operations, monitored automated workflow execution logs, and enforced strict SLA compliance.

KEY TECHNICAL ARCHITECTURE PROJECTS

- **AI MicroMind App & Report Bot Framework:** Natural language to SQL/OData reporting engine across Oracle ERP, SAP, Dynamics 365, and SQL databases.
- **End-to-End Business Automation Pipelines:** Automations across Make, Zapier, and n8n connecting forms, HubSpot/Airtable, email sequences, and Slack.
- **Core Node Architecture (30+ Native Nodes):** Custom Node.js native nodes extending platform capabilities for WhatsApp, Telegram, Gmail, and MCP.
- **Cloud DevOps & Multi-Agent Graphs:** Dockerized multi-agent LangChain/LangGraph pipelines deployed on AWS, DigitalOcean VPS, and Render.

EDUCATION

B.Sc. in Computer Engineering — Faculty of Engineering Shoubra, Benha University
Egypt